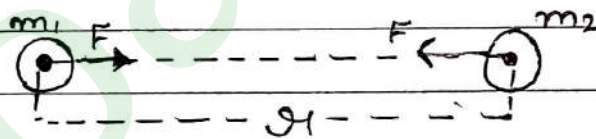


Gravitation

- Gravitation is the Force of attraction b/w any two bodies in the universe.
- It is the weakest Force in nature.
- Gravity: Force of attraction between earth and any object lying on or near its surface.

Newton's universal law of Gravitation

It states that any particle of matter in the universes attracts any other with a force varying directly as the product of masses and inversely as the square of the distance b/w them.



$$F \propto m_1 m_2$$

$$F \propto \frac{1}{r^2}$$

$$F \propto \frac{m_1 m_2}{r^2}$$

$$F = G \cdot \frac{m_1 m_2}{r^2}$$

G - universal gravitational constant

→ G is a force b/w two bodies of unit mass each and placed unit distance apart.

→ Dimension of $G = [M^{-1} L^3 T^{-2}]$

→ Unit of $G = \text{Nm}^2 \text{kg}^{-2}$ or $\text{dyn cm}^2 \text{g}^{-2}$
(SI) (CGS)

→ Value of $G = 6.67 \times 10^{-11} \text{Nm}^2 \text{kg}^{-2}$ or
 $6.67 \times 10^{-8} \text{dyn cm}^2 \text{g}^{-2}$

Feature of Gravitational force

(i) Independent of medium.

(ii) It obeys Newton's third law, $\vec{F}_{12} = -\vec{F}_{21}$

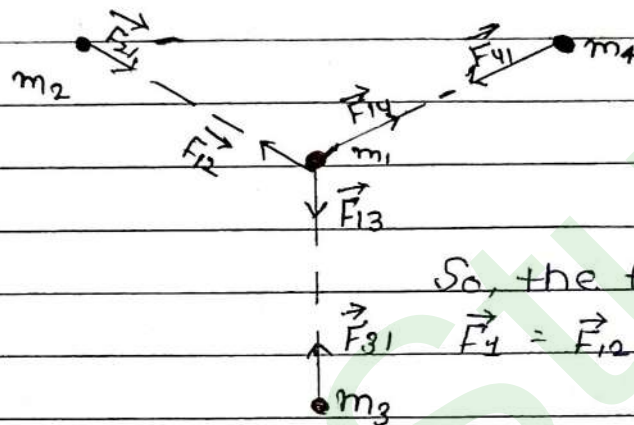
(iii) Valid for point masses.

(iv) Central force and conservative force.

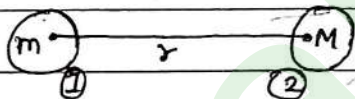
(v) attractive force.

Principles of Superposition of force

- The resultant gravitational force acting on a particle due to a number of masses is the vector sum of all individual force.



- In a regular polygon which have equal masses placed on all of its corners. Then, F_{net} on centre = 0.



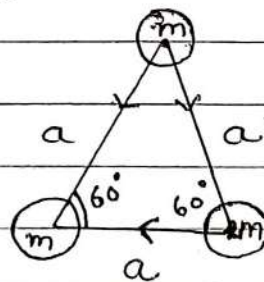
$$\vec{F}_{12} = -\vec{F}_{21}$$

$$\vec{a}_{\text{com}} = 0$$

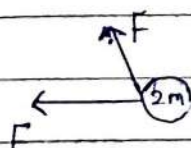
$$\vec{F}_{\text{net}} = 0$$

$$\vec{v}_{\text{com}} = \text{constant}$$

Ques 1. Find net force on Mass 'm' and '2m' if $F_{1m} = \frac{x G m^2}{a^2}$ and $F_{2m} = \frac{y G m^2}{a^2}$. also Find a^2 x and y.



Solⁿ



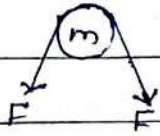
$$F_{\text{net}} = \sqrt{F^2 + F^2 + 2FF \cos 60^\circ}$$

$$= \sqrt{2F^2 + 2F^2 \times \frac{1}{2}} = \sqrt{2F^2 + F^2}$$

$$= F\sqrt{3}$$

$$F_{2m} = \frac{y 2G m^2}{a^2}$$

$$F_{2m} = \frac{\sqrt{3} \cdot 2Gm^2}{a^2} \Rightarrow y = 2\sqrt{3}$$



$$F_{net} = \sqrt{F^2 + (2F)^2 + 2F(2F)\cos 60^\circ}$$

$$= \sqrt{F^2 + 4F^2 + \frac{4F^2 \times 1}{2}}$$

$$= \sqrt{7F^2}$$

$$= \sqrt{7}F$$

$$F_{net} = \sqrt{7} \frac{Gm^2}{a^2} \Rightarrow \boxed{x = \sqrt{7}}$$

(m)

Ques what is the value of x , for which if any body is placed at O , it will experience (F_{net}) gravitational $= 0$?



Soln IF $net = 0$

$$\frac{GMm}{x^2} = \frac{G(3M)m}{(8-x)^2}$$

On solving, $x = \frac{8}{(1+\sqrt{3})} m$

Gravitational Field

→ The space surrounding a material body within which its gravitational force of attraction can be experienced is called its gravitational field.

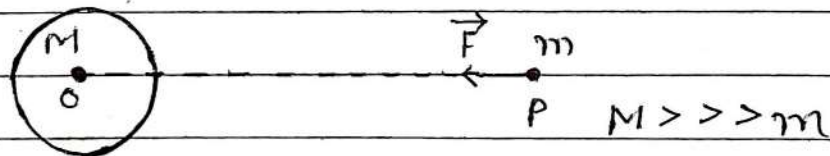
→ e.g Any body brought in earth's gravitational field experiences a force towards its centre.

Intensity of Gravitational Field

→ It is defined as force experienced by unit mass placed at that point provided the presence of unit mass does not disturb the original gravitational field.

→ Denoted by $\vec{E}$

→ Always act towards the mass producing gravitational field.



$$\vec{E} = \frac{\vec{F}}{m}$$

→ Direction of $\vec{E}$ is same as that of $\vec{F}$

$$\left[E = -\frac{dU}{dx} \right]$$

$$F = \frac{GMm}{x^2}$$

$$E = \frac{F}{m} \Rightarrow \boxed{E = \frac{GM}{x^2}}$$
 Sometimes it is written with -ve sign which represents attractive nature.

$$E = \frac{F}{m} = a$$

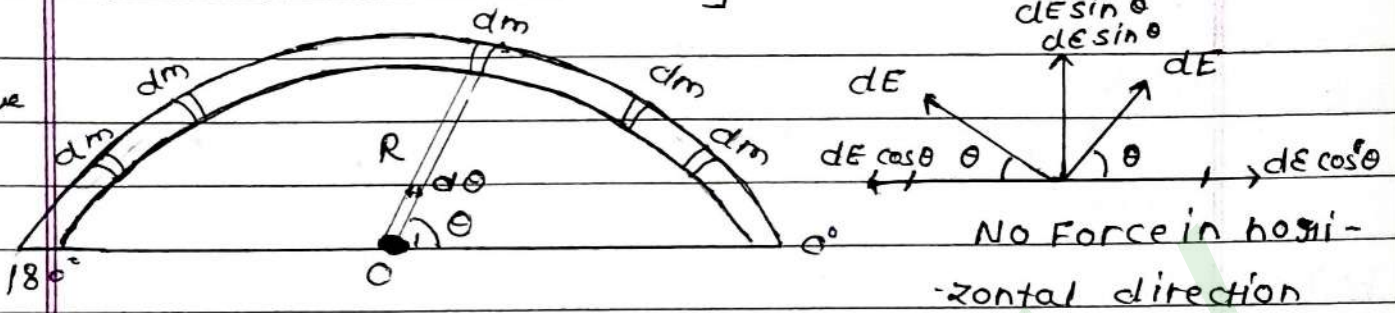
→ The intensity of gravitational field at any point of the earth is equal to the Free acceleration produced in the test mass when placed at that point.

$$\text{Unit of } \vec{E} = \text{Nkg}^{-1}$$

$$\text{Dimension} = [LT^{-2}]$$

Q-2 Find $\vec{E}$ of half ring at centre.

Solve



$\vec{E} = \frac{GM}{r^2}$ is only valid For point masses, to take dm mass.

$$\vec{E} = \int_0^\pi dE \sin \theta + dE \sin \theta + \dots$$

$$= \int_0^\pi \frac{G dm \sin \theta}{R^2}$$

$$= \frac{G}{R^2} \int_0^\pi dm \sin \theta$$

$$= \frac{G}{R^2} \int_0^\pi \frac{m d\theta \sin \theta}{\pi}$$

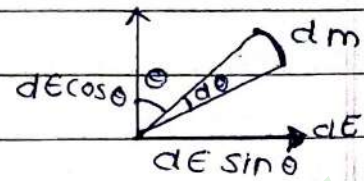
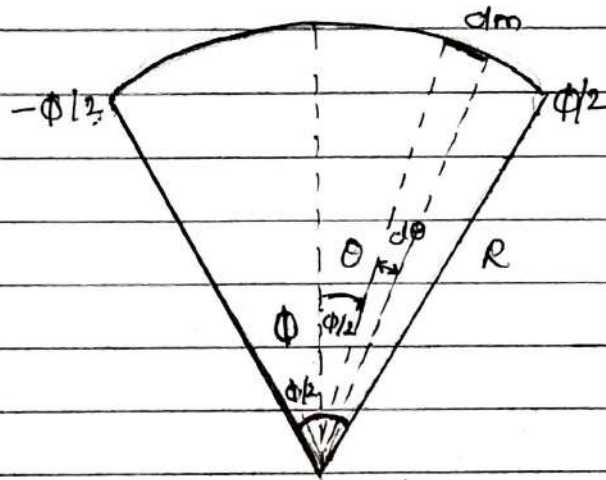
$$= \frac{GM}{R^2} \int_0^\pi \frac{M d\theta \sin \theta}{\pi}$$

$$= \frac{GM}{R^2 \pi} \int_0^\pi \sin \theta d\theta \quad (\int \sin \theta = -\cos \theta)$$

$$= \frac{GM}{R^2 \pi} [-\cos \theta]_0^\pi = -\frac{GM}{R^2 \pi} (-1 - 1)$$

$$\boxed{\vec{E} = \frac{2 GM}{\pi R^2} \hat{j}}$$

Q-3. Find $\vec{E}$ of the given fig.



$\rightarrow dE \sin \theta$ will be cancelled

$$\vec{E} = \int dE \cos \theta$$

$$= \int_{-\phi/2}^{\phi/2} \frac{G M}{R^2} dm \cos \theta = \frac{G}{R^2} \int dm \cos \theta$$

$$= \frac{G}{R^2} \int_{-\phi/2}^{\phi/2} \frac{m d\theta}{\phi} \cos \theta$$

$$\frac{M}{R\phi} = \frac{dm}{R d\theta}$$

$$dm = \frac{m d\theta}{\phi}$$

($\phi = \text{constant}$)
Given

$$= \frac{G M}{R^2 \phi} \int_{-\phi/2}^{\phi/2} (\cos \theta) d\theta = \frac{G M}{R^2 \phi} [\sin \theta]_{-\phi/2}^{\phi/2}$$

$$= \frac{G M}{R^2 \phi} \left[\frac{\sin \phi}{2} - \sin \left[-\frac{\phi}{2} \right] \right]$$

$$\boxed{\vec{E} = \frac{G M}{R^2 \phi} \left[\frac{2 \sin \frac{\phi}{2}}{2} \right]} \quad \text{General formula}$$

$\downarrow$

compare with Half Ring -

$$E = \frac{G M}{R^2 \pi} \cdot \frac{2 \left(\sin \frac{\pi}{2} \right)}{2} = \frac{2 G M}{\pi R^2} \hat{j}$$

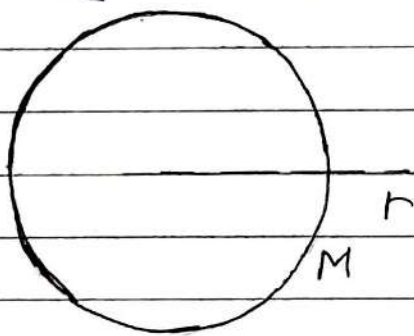
$\sin 90^\circ = 1$

Compare with quadrant -

$$E = \frac{G M}{R^2 \pi/2} \cdot \frac{2 \sin 90^\circ}{2} = \frac{2 G M}{R^2 \pi} \cdot \frac{2 \times 1}{\sqrt{2}} = \frac{2 G M \sqrt{2}}{R^2 \pi}$$

Shell Theorem

It states that a spherically symmetric mass exerts no net gravitational force on an object located inside it, and it can be treated as if all its mass is concentrated at its center for objects located outside.



$\leftarrow F$ m

r

M

Shell 'feels like' a point mass from outside

$$F = \frac{GMm}{r^2}$$

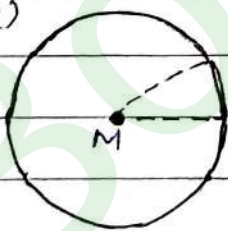
M

outside.

$\leftarrow$ m

Gravitational Field of Hollow sphere (shell)

(M, R)



$\leftarrow E$ P

$$E = \frac{GM}{r^2}$$

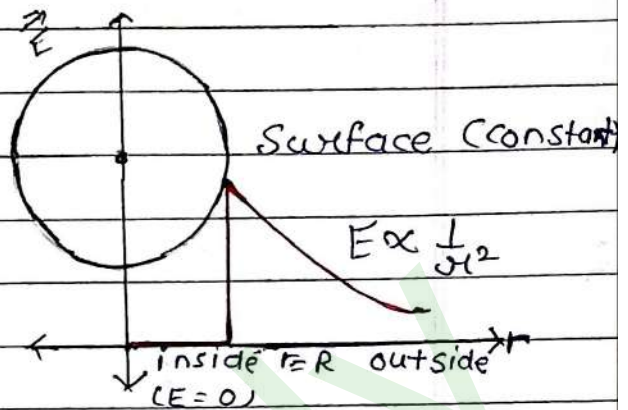
→ Uniform mass distribution (symmetric)

→ mass lies on centre.

Graph

(i) Outside [$x > R$]

$$E = \frac{GM}{x^2} \Rightarrow E \propto \frac{1}{x^2}$$

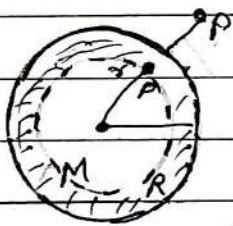
(ii) At Surface [$x = R$]

$$E = \frac{GM}{R^2} \Rightarrow E \propto \frac{1}{R^2}$$

(iii) Inside [$x < R$]

$$E = 0$$

- Gravitational Field of solid sphere

(i) outside [$x > R$]

$$E = \frac{GM}{R^2}$$

$$E = \frac{GM}{R^2}$$

(ii) Inside [$R > x$]

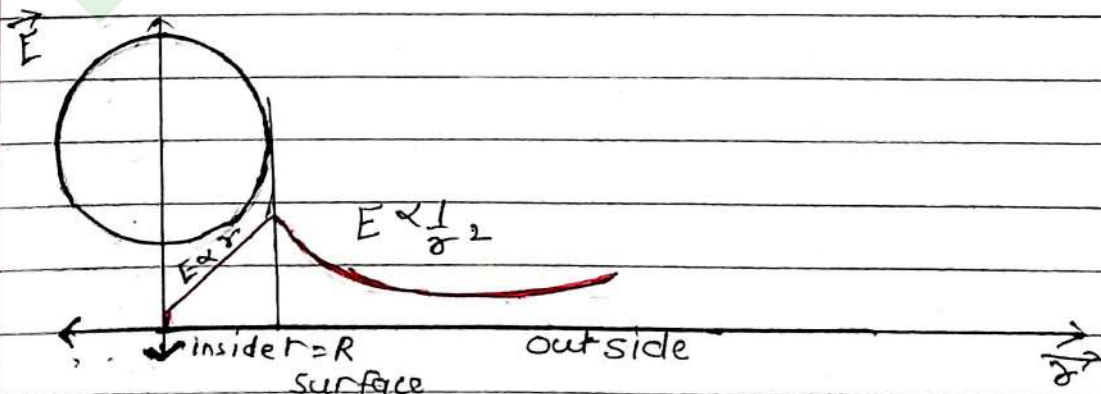
$$\vec{E}_P = \vec{E}_{\text{shell inside point}} + \vec{E}_{\text{solid sphere surface point}} \quad [m' = \rho \times V]$$

$$\vec{E} = \frac{G\rho V}{x^2} = \frac{G\rho \frac{4}{3}\pi x^3}{x^2} = \frac{4G\rho\pi x}{3}$$

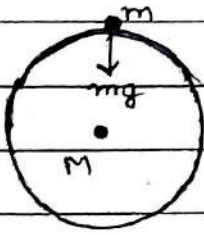
$$\vec{E} = \frac{4G\rho\pi x}{3} \text{ or } \frac{GM}{R^3}$$

[At centre, $E = 0$]

Graph

 $E \propto x$ $\downarrow$ g_{surface} 

Acceleration due to gravity of Earth -



$$F = \frac{GmM}{r^2}$$

$$mg = \frac{GmM}{r^2}$$

$$g = \frac{GMe}{r^2}$$

→ g is independent of m and same for all types of mass.

At surface, $r = R_e$

$$g = \frac{GMe}{R_e^2}$$

$$M_e = 5.97 \times 10^{24} \text{ kg} \approx 6 \times 10^{24} \text{ kg}$$

$$R_e \approx 6.4 \times 10^6 \text{ m}$$

$$G = 6.67 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$$

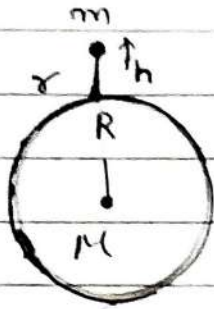
$$g = \frac{6.67 \times 10^{-11} \times 6 \times 10^{24}}{(6.4 \times 10^6)^2} \approx 9.8$$

$$g = 9.8 \text{ m/s}^2$$

→ Proportionality in term of $M \Rightarrow g \propto M/R^2$

Proportionality in terms of density $\Rightarrow g \propto \rho R$

Variation of g with Altitude (above Earth's Surface)



$$r = h + R$$

$$g' = \frac{GM_e}{R^2} = \frac{GM_e}{(R+h)^2} = \left[g = \frac{GM_e}{R^2} \Rightarrow GM_e = gR^2 \right]$$

$$g' = \frac{gR^2}{(R+h)^2}$$

$$= \frac{g}{\left[\frac{R+h}{R} \right]^2}$$

Learn

$$g' = \frac{g}{\left(1 + \frac{h}{R} \right)^2}$$

$$g' = \frac{g \cdot R^2}{(R+h)^2}$$

Binomial Theorem

$$\text{If } h \ll R$$

$$\left\{ \begin{aligned} (1+x)^{-n} &\ll 1 \\ &\approx 1 - nx \end{aligned} \right\}$$

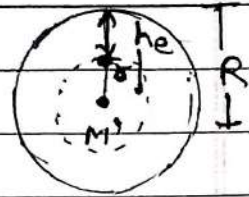
$$g' = \frac{g \left[1 + \frac{h}{R} \right]^{-2}}{R^2} = \frac{g \left[1 - \frac{2h}{R} \right]}{R^2}$$

Learn

Variation of g with depth

$M = \text{Volume} \times \text{density}$

$$M' = \frac{4\pi R^3 \rho}{3}$$



$$[r = R - h]$$

$$g' = \frac{G \times \frac{4\pi R^3 \rho}{3}}{r^2} = \frac{4\pi G R \rho}{3}$$

$$\boxed{g' = G \rho \frac{4\pi R}{3}}$$

$$\rho = \frac{M_e}{\frac{4\pi R_e^3}{3}}$$

$$g' = \frac{G M_e \times \frac{4\pi R}{3}}{\frac{4\pi R \times R_e^3}{3}}$$

$$\star \boxed{g' = \frac{G M_e R}{R_e^3}} \quad g' \propto R$$

In terms of height

$$g' = \frac{G M_e r}{R_e^3} \quad (r = R_e - h)$$

$$g' = \frac{g R_e^2 R}{R_e^3}$$

$$(g = \frac{G M_e}{R^2} \Rightarrow G M_e = g R^2)$$

$$\boxed{g' = \left[\frac{g}{R_e} \right] r}$$

$$\Rightarrow \boxed{g_d = g \left[1 - \frac{h}{R_e} \right]} \quad h = \text{depth, } d.$$

* At centre, $g = 0$

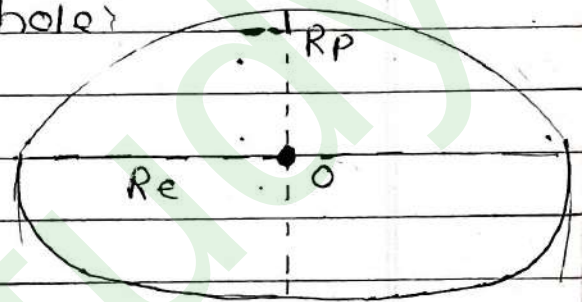
Variation of g with shape of Earth

- Earth is not a perfect sphere (orange shape)

R at quarter $> R$ at pole

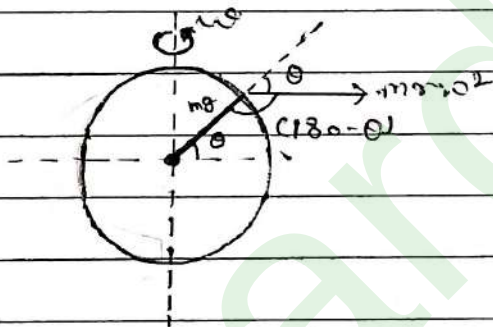
So,

$$g_e < g_p \quad (g \propto \frac{1}{R^2})$$



Variation of g with Latitude

(Rotation of Earth)

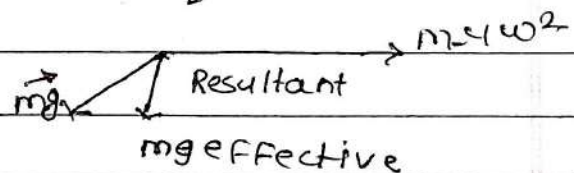


g decreases due to $m\omega^2 r$

★ Difference is very small.

[θ = Latitude]

(angle measure from equator)



→ mg effective is not along centre of Earth.

$$mg_{\text{eff}} = m\vec{g} + m\omega^2 r$$

$$[\vec{g}_{\text{eff}} = \vec{g} + \vec{a}\omega^2]$$

$$g_{\text{eff}} = \sqrt{(g)^2 + (\cancel{r\omega^2})^2 + 2gr\omega^2 \cos(180^\circ - \theta)}$$

ω is very small

$$= \sqrt{g^2 - 2gr\omega^2 \cos \theta}$$

$$= g \sqrt{1 - \frac{2r\omega^2 \cos \theta}{g}} \quad (\text{multiply and divide by } g)$$

$$= g \left[\underbrace{1 - \frac{2r\omega^2 \cos \theta}{g}}_{\text{very small}} \right] \quad [(1-x)^n = 1 - nx \quad (x \ll 1)]$$

$$g_{\text{eff}} = g \left[1 - \frac{1}{2} \frac{2r\omega^2 \cos \theta}{g} \right]$$

$$g_{\text{eff}} = g - r\omega^2 \cos \theta$$

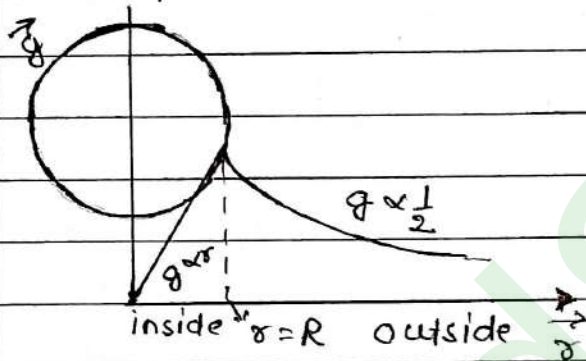
At equator, $\theta = 0$

$$g' = g - R\omega^2 \quad (\text{minimum})$$

At pole, $\theta = 90^\circ$

$$g' = g \quad (\text{maximum})$$

Graph



Que

Find weight of a person at depth R below earth's surface if weight at surface is 100 N.

Solⁿ

$$g' = g \left[1 - \frac{R}{2R} \right] = g/2$$

$$w = \frac{mg}{2} = 50 \text{ N}$$

Que what should be the speed of rotation and time period of earth so that the weight of a person at equator is zero?

Ans

$$\begin{aligned} \text{At equator, } g' &= g - R\omega^2 \\ 0 &= g - R\omega^2 \\ g &= R\omega^2 \\ \left[\omega &= \sqrt{\frac{g}{R}} \right] \end{aligned}$$

$$\begin{aligned} T &= \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{R}{g}} \\ T &= 2\pi \sqrt{\frac{R}{g}} \approx 84 \text{ minute.} \end{aligned}$$

Gravitation Potential (V)

→ It is defined as the amount of work done in bringing a body of unit mass from infinity to that point.

Potential difference



$$V_A - V_B = \frac{W_{B \rightarrow A}}{m}$$

$$V_A - \infty = \frac{W_{\infty \rightarrow A}}{m} \quad \text{ext. agent.}$$

$$\boxed{V_A = \frac{W}{M}}$$

→ Condition - mass should be moved very slowly ($\Delta KE = 0$)

→ Scalar Quantity.

V_1 due to a mass 'm'

$$V_p = -\frac{GM}{r}$$

→ Gravitation potential is always come to be -ve.

Relationship b/w Gravitational field & Potential.

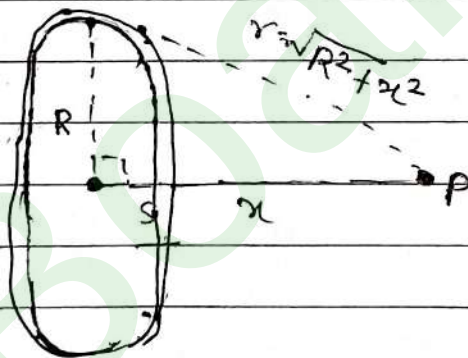
$$E = -\frac{dV}{dr}$$

→ towards M

$$E_x = -\frac{\partial V}{\partial x}, E_y = -\frac{\partial V}{\partial y}, E_z = -\frac{\partial V}{\partial z}$$

Exe

Find v_p of the ring
dm (point mass)



extended mass

$$\int dV = \int -\frac{G dm}{r}$$

$$V_p = -\frac{G}{\sqrt{R^2 + x^2}} \int dm$$

$$\left[V_p = -\frac{GM}{\sqrt{R^2 + x^2}} \right] \text{ Axis}$$

IF $x \gg R$

$$\left[V_p = -\frac{GM}{x} \right]$$

$$- V_p \text{ at centre} = 0 \cdot -\frac{GM}{R} \quad (x=0)$$

$$E = -\frac{dV}{dx} = d \left[\frac{GM}{\sqrt{R^2 + x^2}} \right]$$

$$= GM d (R^2 + x^2)^{-1/2}$$

$$= GM \left[\frac{-1}{2} \right] (R^2 + x^2)^{-\frac{1}{2}-1}$$

$$= GM \left[\frac{-1}{2} \right] (R^2 + x^2)^{-3/2} \times (0 + 2x)$$

chain Rule

$$E_x = -\frac{GMx}{[R^2 + x^2]^{3/2}}$$

V_p due to hollow sphere

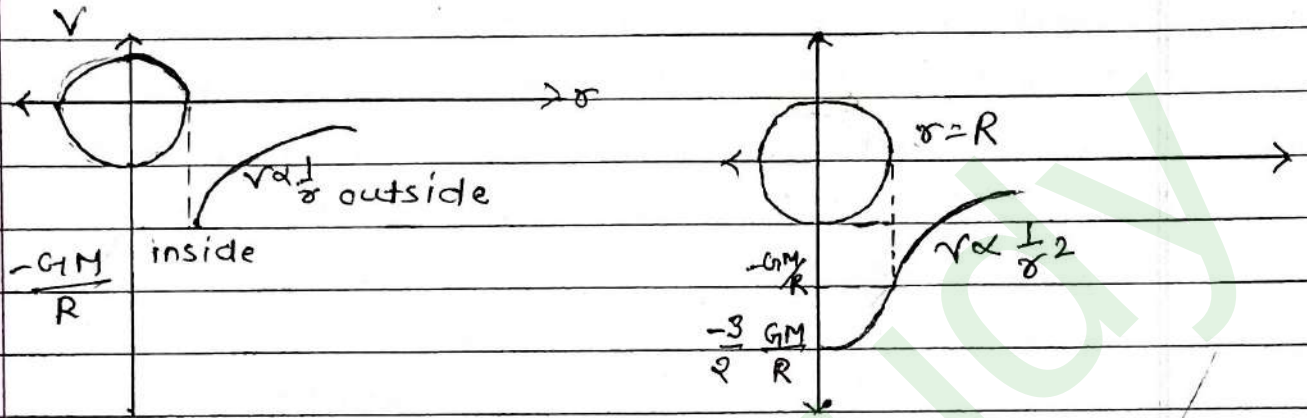
	Outside ($r > R$)	Surface ($r = R$)	Inside ($r < R$)
E	$\propto \frac{GM}{r^2}$	$\propto \frac{GM}{R^2}$	0
V	$-GM/r$	$-GM/R$	$-GM/R$

Solid sphere

	Outside	Surface	Inside
E	$\propto \frac{GM}{r^2}$	$\propto \frac{GM}{R^2}$	$\propto \frac{GM}{R^3}$
V	$-GM/r$	$-GM/R$	$-\frac{GM}{R^3} \left[\frac{3R^2 - r^2}{2} \right]$

Graph

(i) Hollow sphere



Ques Gravitation potential at centre of a solid sphere $V_0 = ?$

Centre = inside

$$V_i = -\frac{GM}{R^3} \left[\frac{3R^2}{2} - \frac{1r^2}{2} \right] \quad (r=0)$$

$$= -\frac{GM}{R^3} \left[\frac{3}{2} R^2 \right]$$

$$= -\frac{3}{2} \frac{GM}{R}$$

$$= V_0 = \frac{3}{2} V_s$$

Gravitational Potential Energy (U)

- Energy associated with a body due to its position in the gravitational field of another body and is measured by the amount of work done in bringing a body from infinity to a given point in the gravitational field of the other.

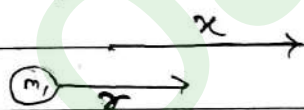
- U is not defined for a single mass
- Scalar quantity.
- Always comes to be negative.

$$\Delta U = -W_{\text{conservative internal force}}$$

$$\text{At } \infty, U_{\infty} = 0$$

$U_g - \cancel{U_{\infty}} = -W_{\text{gravitational force in any configuration}}$ bringing each mass from ∞

$$W_{g1} = 0 \text{ (single mass)}$$



$$F_g = \frac{G m_1 m_2}{x^2}$$

variable.

$$W_{g2} = \int_{\infty}^x f dx = \int_{\infty}^x -\frac{G M_1 M_2}{x^2} dx$$

$$= -G m_1 m_2 \int_{\infty}^x \frac{dx}{x^2}$$

$$= -G m_1 m_2 \left[-\frac{1}{x} \right]_{\infty}^x$$

$$W_{g2} = \frac{G m_1 m_2}{x}, \quad U = -W_g$$

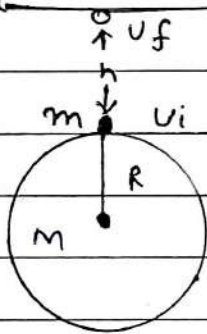
$$U = -\frac{G m_1 m_2}{x}$$

$$[\Delta U = -W_g = +W_{ext}]$$

→ IF m is moved from x_1 to x_2

$$\Delta U = \int_{x_1}^{x_2} \frac{GMm}{x^2} dx = GMm \left[\frac{-1}{x} \right]_{x_1}^{x_2}$$

$$\Delta U = GMm \left[\frac{1}{r_1} - \frac{1}{r_2} \right]$$



IF a body is moved from a surface of earth to a point at height h , then change in its gravitation P.E will be :-

(work done = $\Delta U = U_f - U_i$)

$$U_i = -\frac{GmM}{R}, \quad U_f = -\frac{GmM}{(R+h)}$$

$$\Delta U = U_f - U_i$$

$$= -\frac{GmM}{R+h} - \left[-\frac{GmM}{R} \right]$$

$$= GmM \left[\frac{1}{R} - \frac{1}{R+h} \right]$$

$$= GmM \left[\frac{R+h-R}{R(R+h)} \right] \quad (GM = gR^2)$$

$$= \frac{GmMh}{R(R+h)}$$

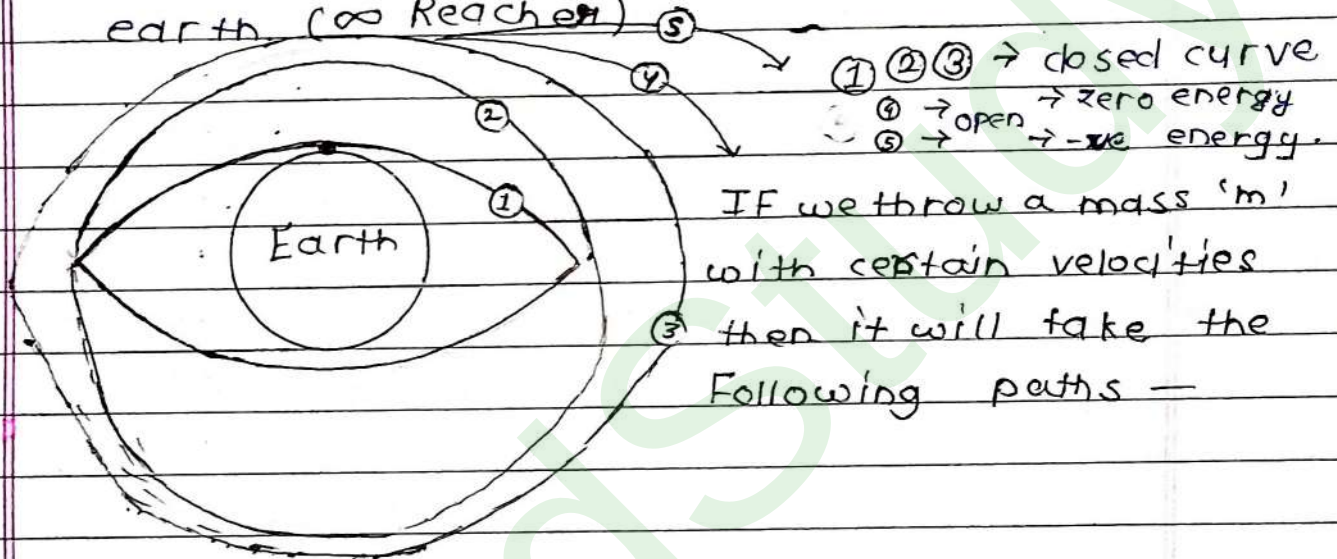
$$= \frac{gR^2mh}{R(R+h)} = \frac{mgh}{\left(1 + \frac{h}{R}\right)} = \boxed{\Delta U = mgh \left(1 + \frac{h}{R}\right)}$$

In day to day life, $h \ll R$

$$\boxed{\Delta U = mgh}$$

Escape Velocity (v_e)

* Escape Velocity is the minimum velocity with which a body must be projected vertically upwards in order that it escape the gravitational field of earth (∞ Reaches)



1. velocity ($\uparrow\uparrow$) $\rightarrow$ elliptical path
2. $v = \sqrt{gR} \approx 8 \text{ km/s} \rightarrow$ circular path.
3. $v > \sqrt{gR} \rightarrow$ elliptical path.
4. $v = \sqrt{2gR} \approx 11.2 \text{ km/s} \rightarrow$ Parabolic path $\rightarrow$ Escape velocity.
5. $v > \sqrt{2gR} \rightarrow$ Hyperbolic path $\rightarrow$ Escape velocity (v_e)

Conservation of mechanical Energy

$$U_i + K_i = U_f + K_f \quad (\text{use for numericals})$$

$$-\frac{GMm}{R} + \frac{1}{2}mv_e^2 = 0 \text{ to } \infty$$

$$\frac{1}{2}mv_e^2 = \frac{GMm}{R}$$

$$v_e \propto \sqrt{\frac{m}{R}}$$

$$v_e \propto R^{-\frac{1}{2}}$$

$$V_e = \sqrt{\frac{2GM}{R}} \leftarrow \text{For any planet}$$

$$V_e = \sqrt{\frac{2gR^2}{R}} \Rightarrow \boxed{V_e = \sqrt{2gR}} \leftarrow \text{For earth}$$

$$\approx 11.2 \text{ km/s.}$$

If we gave velocity more than escape velocity then the particle have same remaining speed out of gravitational field $\rightarrow$ inter scalar speed.

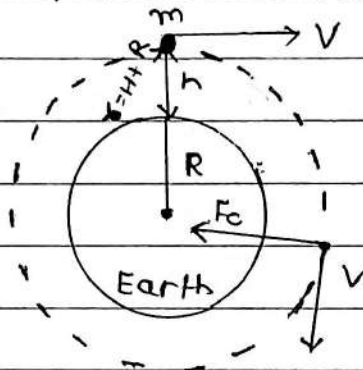
$\rightarrow$ Escape velocity does not depend on the mass of a body.

Heavenly body	Escape velocity
moon	2.3 km/s
mercury	4.28 km/s
Earth	11.2 km/s
Jupiter	60 km/s
Sun	618 km/s
Neutron star	$2 \times 10^5 \text{ km/s}$

$\rightarrow$ Escape velocity of moon is less than its root mean velocity of its atmospheric molecules. That is why, moon has no atmosphere.

Orbital Velocity (V_o)

→ It is the velocity required to put the satellite into its orbit around the earth.



$$F_c = \frac{mv^2}{r} \quad \text{--- (I)}$$

$$F_g = \frac{GmM}{r^2} \quad \text{--- (II)}$$

$$F_c = F_g$$

$$\frac{mv^2}{r} = \frac{GmM}{r^2}$$

$$v^2 = \frac{GM}{r} \Rightarrow \boxed{v_o = \sqrt{\frac{GM}{r}}} \quad \begin{array}{l} \text{dist. b/w satellite} \\ \text{and centre of} \\ \text{earth.} \end{array}$$

► If satellite is revolving very close to Earth's surface.

$$r \rightarrow R$$

$$v_o = \sqrt{\frac{GM}{R}} = \sqrt{\frac{gR^2}{R}} \Rightarrow \boxed{v_o = \sqrt{gR}} \approx 7.9 \text{ km/s}$$

→ Orbital velocity of a satellite decreases with the increase in the radius of orbit.

* Relation b/w orbital velocity (v_o) & escape velocity (v_e)

$$v_e = \sqrt{2gR} \quad (\text{earth's surface}) \quad \text{--- (I)}$$

$$v_o = \sqrt{gR}$$

Divide (I) and (II) $\frac{v_e}{v_o} = \frac{\sqrt{v_e}}{\sqrt{v_o}} = \frac{\sqrt{2gR}}{\sqrt{gR}} \quad \begin{array}{l} \text{orbital} \\ \text{velocity in} \\ \text{terms of } g \end{array}$

Difference b/w v_e and v_o for earth = 3.4 km/s.

$$v_o = \sqrt{\frac{gR}{(1+h/R)}}$$

Time Period of a Satellite

- It is the time taken by a satellite to complete one revolution around the earth.

$$T = \frac{\text{Circumference of orbit}}{\text{Orbital velocity}} = \frac{2\pi r}{V_0}$$

$$T = \frac{2\pi r}{\sqrt{GM}} = \frac{2\pi r^{1+\frac{1}{2}}}{\sqrt{GM}}$$

$$\boxed{T = \frac{2\pi r^{3/2}}{\sqrt{GM}}} \quad (r = \text{dist b/w centre of earth \& satellite})$$

- T when satellite is very close to Earth's surface :-

$$T = \frac{2\pi R}{\sqrt{\frac{GM}{R}}} = \frac{2\pi R}{\sqrt{gR}} \Rightarrow \boxed{T = 2\pi \sqrt{\frac{R}{g}}} \approx 84.6 \text{ min.}$$

$$\text{Binding Energy} = T \cdot E = KE$$

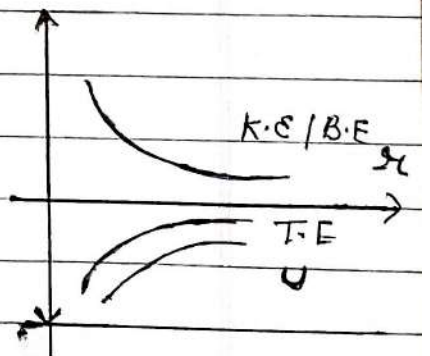
↓

(min. energy required to take particle out ∞)

Energy of satellites

$$KE = \frac{1}{2}mv^2 = \frac{1}{2}m \left[\sqrt{\frac{GM}{r}} \right]^2 = \frac{GMm}{2r}$$

$$U = -\frac{GMm}{r}$$



$$\text{Total Energy} = KE + P.E \Rightarrow$$

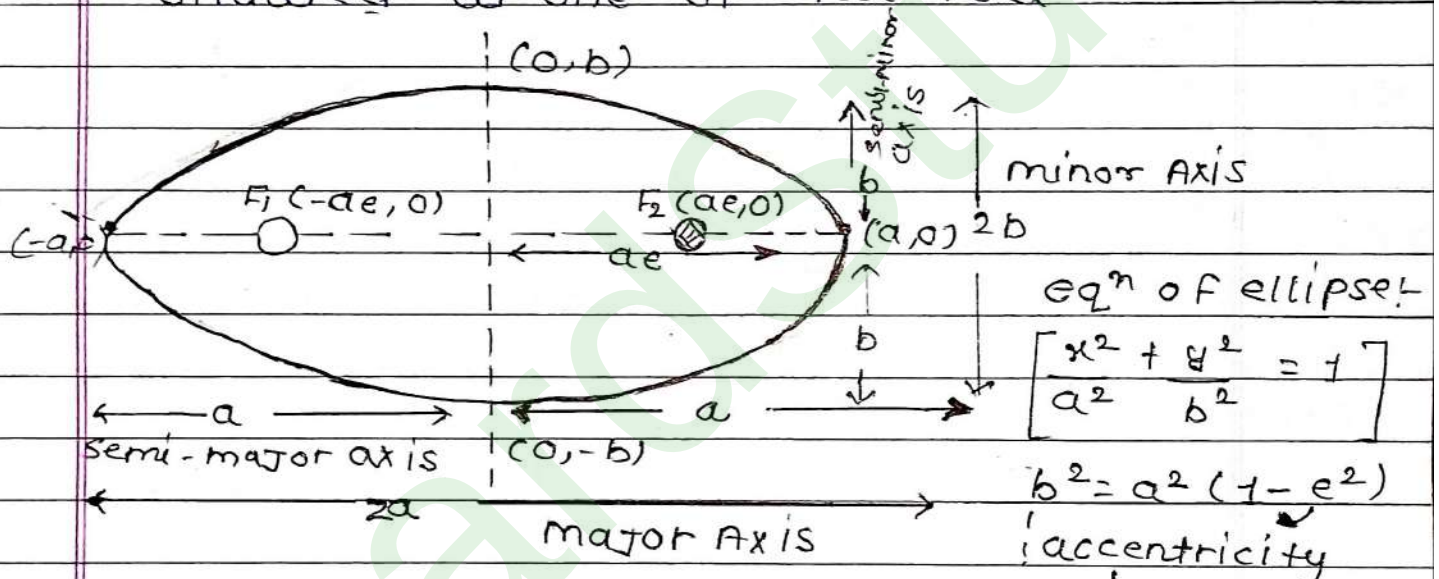
$$\boxed{TE = -\frac{GMm}{2r}}$$

Kepler's law of Planetary Motion

→ To explain motion of planets. Kepler Formulated three laws -

* First law (law of orbits)

"Each planet revolves around the sun in an elliptical orbit with sun situated at one of two foci."



→ closed distance b/w Sun & planet (or $e < 1$)
Perihelion OR Perigee [$r = a(1 - e)$]
min

→ Farthest distance - Aphelion OR Aphogee
[$r = a(1 + e)$] * $\left[\frac{2}{r} = \frac{1}{r_{\max}} + \frac{1}{r_{\min}} \right]$

* Second law (Law of Area)

- Area swept by a planet is equal in equal interval of time around the Sun.

"The areal velocity of a planet is constant"

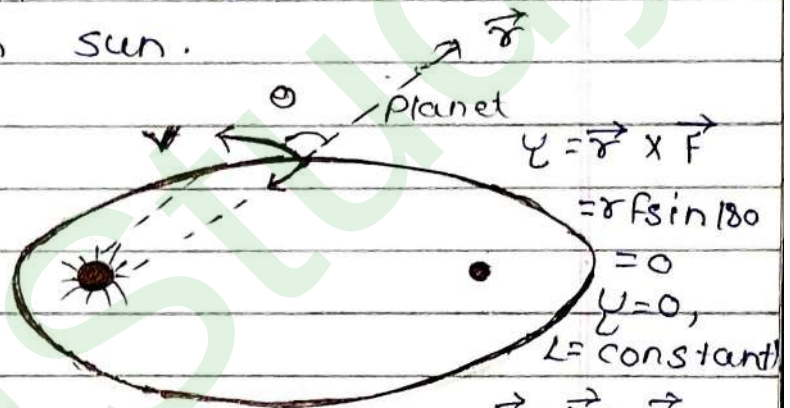
$$\frac{dA}{dt} = \text{Constant}$$

→ Planet do not revolve around the sun with constant speed.

→ The moves faster when they are near to sun & slower when farther from sun.

$$\frac{A_1}{t_1} = \frac{A_2}{t_2} = \frac{A_3}{t_3}$$

proof of second law



planet move in time, dt

distance covered

$$= v dt$$

$$\sin(180^\circ - \theta) = \frac{h}{v dt}$$

$[h = v dt \sin \theta]$

$$\vec{L} = \vec{r} \times \vec{p}$$

$$= \vec{r} \times m \vec{v}$$

$$[L = m v r \sin \theta]$$

$$\text{Area} = \frac{1}{2} \times b \times h$$

(constant)

$$m_1 v_1 r_1 = m_2 v_2 r_2$$

$$\frac{dA}{dt} = \frac{1}{2} \times r \times v dt \sin \theta \quad \text{--- (i)}$$

Multiply and divide by m

$$\frac{dA}{dt} = \frac{1}{2} \frac{m v r \sin \theta}{m} dt$$

$\frac{dA}{dt} = \frac{L}{2m}$	$= \text{constant}$
	$\leftarrow \text{mass of planet}$

Hence, proved

Speed of planet at Aphelion & Perihelion

→ Angular momentum of planet is constant ②

$L_1 = L_2$
 $r_1 \times p_1 = r_2 \times p_2$
 $r_1 m v_1 \sin 90^\circ = r_2 m v_2 \sin 90^\circ$
 $v_1 r_1 = v_2 r_2$
 $v_1 a(1+e) = v_2 a(1-e) \quad \text{--- (a)}$

Conservation of Energy

$$U_1 + K_1 = U_2 + K_2$$

$$-\frac{GSM}{r_1} + \frac{1}{2} m v_1^2 = -\frac{GSM}{r_2} + \frac{1}{2} m v_2^2 \quad \text{--- (b)}$$

Putting value of eq.ⁿ (a) in (b) we get

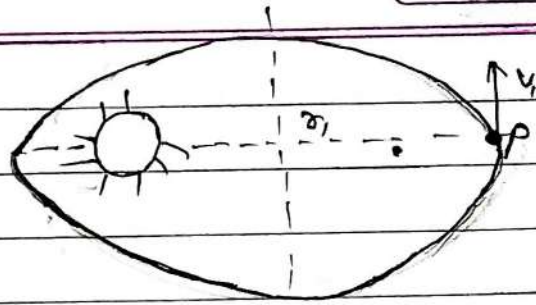
$V_1 = \sqrt{\frac{GMS(1-e)}{a(1+e)}} \quad \text{Aphagee}$	$V_2 = \sqrt{\frac{GMS(1+e)}{a(1-e)}} \quad \text{Perigee}$
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* Third law (Law of Periods)

- 'Square of period of revolution of a planet around the sun is proportional to the cube of the semimajor axis of its elliptical orbit.'

$$T^2 \propto a^3 \quad \left[\frac{T_2}{T_1} \right]^2 = \left[\frac{a_2}{a_1} \right] \rightarrow \text{used in numericals.}$$

Proof



$$\frac{dA}{dt} = \frac{L}{2m}$$

1 complete revolution, Area swept = Area of ellipse = πab

$$\text{time} = T$$

$$\frac{\pi ab}{T} = \frac{L}{2m} \quad (L = r_1 m v_1 \sin 90^\circ = r_1 m v_1)$$

$$\frac{\pi ab}{T} = \frac{r_1 m v_1}{2m}$$

$$\frac{\pi ab}{T} = a(1+e) \sqrt{\frac{GM_s(1-e)}{a(1+e)}}$$

Squaring both sides:-

$$\frac{\pi^2 b^2}{T^2} = (1+e)^2 \frac{GM_s(1-e)}{a(1+e)}$$

$$\frac{\pi^2 a^2 (1-e^2)}{T^2} = (1+e)^2 \frac{GM_s(1-e)}{a(1+e)} \quad (b^2 = a^2(1-e^2))$$

$$\frac{2\pi^2 a^3}{T^2} = GM_s$$

$$T^2 = \frac{2\pi^2 a^3}{GM_s} \text{ constant.}$$

$[T^2 \propto a^3]$ Hence proved.

Ques what will be the distance b/w the planet and earth whose time period is 64 times as that of Earth? ($R_{\text{earth}} = R$)

Sol- $\left(\frac{r_1}{r_2}\right)^3 \propto \left(\frac{T_1}{T_2}\right)^2 \Rightarrow \left(\frac{R}{r_2}\right)^3 \propto \left(\frac{T}{64T}\right)^2$

Ques If the masses are tripled and the distance b/w them is doubled, the $\vec{F}_g$ b/w them will become?



$$F' = \frac{G(3m)(3m)}{2d} = \frac{9Gm^2}{4d^2} = \frac{9F}{4}$$